PCT INTERNATIONAL APPLICATION

TITLE OF INVENTION

AETHYR ONE: AN INTEGRATED INDUSTRIAL PACKAGE FOR ORBITAL ENERGY HARVESTING, EXOTIC MATTER PRODUCTION, HIERARCHICAL COMPUTATION, AND TORQUE-AMPLIFIED PROPULSION

TECHNICAL FIELD

[0001] The present disclosure relates to a unified industrial package, hereafter referred to as "Aethyr One." The invention integrates magnetospheric energy harvesting, photon-driven antimatter production, zetta-scale symbiotic computation, hierarchical neural architectures, and cascaded electromagnetic motors into a single, closed-loop technical ecosystem.

BACKGROUND ART

[0002] Current industrial paradigms are fragmented, treating energy generation, computation, and mechanical propulsion as distinct problems with separate energy costs. Orbital solar power is limited by flux density; antimatter production is limited by energy efficiency; and neural networks are limited by $O(n^2)$ complexity. There is a requirement for a unified "Aethyr One" package that solves these bottlenecks through technological synergy.

SUMMARY OF INVENTION

[0003] Aethyr One is a self-sustaining industrial package comprising four primary interdependent subsystems:

[0004] 1. The Power Subsystem (ORRT): Utilizing Overlapping Ring Resonance and a Lorentz-Gradient Ratchet (LGR) to extract gigawatt-scale electrical power from planetary magnetospheres.

[0005] 2. The Production & Logic Subsystem (ACN/Symbiotic Anticomputer): A modular node that utilizes the ORRT power to generate antimatter. Crucially, it harvests the relativistic "waste" electron stream from the production process to perform zero-marginal-cost Zetta-scale computation.

[0006] 3. The Control Subsystem (HST): A Hierarchical Sequence Transformer neural architecture utilizing Pell-Lucas recursive temporal encoding to manage the massive data and decision-making requirements of the Aethyr One package with $O(n \log n)$ efficiency.

[0007] 4. The Motive Subsystem (Cascade Motor): A torque-multiplied electromagnetic motor utilizing inter-stage capacitor impulses ("kicks") to convert the package's electrical output into high-torque mechanical or propulsive work.

[0008] The Aethyr One package operates as a "Global Brain and Factory," where the byproduct of one stage (e.g., waste electrons) serves as the primary fuel for the next (e.g., zetta-scale computation), creating a near-zero-waste industrial loop.

DETAILED DESCRIPTION

1. The Power Core (ORRT/LGR) [0009] The Aethyr One package derives its primary energy from the interaction of superconducting toroidal rings and the planetary magnetic gradient tensor ($T = \nabla\vec{B}$). By operating at high angular velocities (>1,200 rad/s), the system establishes a Lorentz-Gradient Ratchet. This allows the package to maintain high-capacity electrical output without traditional fuel consumption.
2. The Symbiotic Processor (ACN/Anticomputer) [0010] Integrated within the Aethyr One package is the Anticomputer Node. It utilizes laser-wakefield acceleration to create electron-positron pairs. The "Symbiotic" feature is essential: it captures the 0.99c electron stream produced during pair creation. This stream is passed through Micro-Magnetic Logic Arrays (M^2LA). In this package, the computation is not an independent energy expense; it is a "free" byproduct of fuel production.
3. The Intelligence Layer (HST) [0011] Aethyr One utilizes the Hierarchical Sequence Transformer for system-wide governance. By employing a recursive spine based on the Pell-Lucas sequence ($P(n) = 2P(n-1) + P(n-2)$), the system maintains a compressed, hierarchical memory of all package operations. This prevents the computational "bloat" associated with standard AI, allowing Aethyr One to scale indefinitely across orbital constellations.
4. The Mechanical Drive (Cascade Motor) [0012] To translate electrical energy into work, Aethyr One employs the Cascade Motor. This subsystem utilizes high-voltage capacitors to "kick" successive stages of the motor. It acts as an electronic torque converter, allowing the Aethyr One package to drive heavy industrial loads or propulsion systems with extreme efficiency.

CLAIMS

What is claimed is:

1. The Aethyr One integrated industrial package, comprising: an orbital energy harvesting subsystem (ORRT) utilizing a Lorentz-Gradient Ratchet; an antimatter production and symbiotic computation subsystem (ACN); a hierarchical neural control subsystem (HST) utilizing Pell-Lucas temporal encoding; and a cascaded electromagnetic motor subsystem; wherein these subsystems are integrated into a single closed-loop energy and data environment.
2. The package of claim 1, wherein the ACN subsystem performs zetta-scale computation as a zero-marginal-energy-cost byproduct of the antimatter production process.
3. The package of claim 1, wherein the HST subsystem provides a decentralized operating framework for the ORRT and ACN components.
4. The package of claim 1, wherein the Cascade Motor subsystem multiplies the torque output of the integrated package using inter-stage capacitor-driven impulses.

ABSTRACT

The Aethyr One package is an integrated industrial system comprising magnetospheric energy harvesting (ORRT), symbiotic antimatter and computational production (ACN), hierarchical neural governance (HST), and torque-amplified motive force (Cascade Motor).

The invention functions as a unified "package" for self-sustaining orbital or terrestrial infrastructure.